

Mini Project 2 – Streamlit Chatbot

Introduction

The objective of this project was to design and implement a conversational chatbot using OpenAI's GPT-3.5-Turbo model and the Streamlit framework in Python.

The chatbot maintains conversation history using `st.session_state` and generates responses based on the full conversation context. The system sends the entire message history to the OpenAI API to enable contextual understanding and follow-up interactions.

The chatbot was tested with different types of queries to evaluate:

- Basic conversational ability
- Context retention (memory)
- Technical explanations
- Follow-up questions
- Multi-part (Technical and Non Technical) prompts
- Personal advice

Basic question:

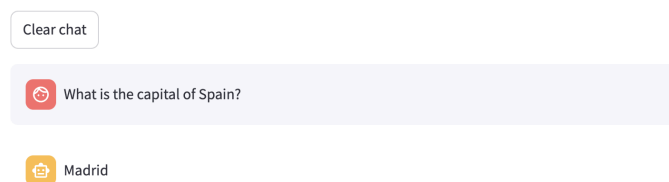
Test Prompt:

What is the capital of Spain?

Observation:

The chatbot correctly responded with "Madrid."

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Conversation Memory Test

Test Prompts:

My name is Priya and my friend name is Lizzy.we both project teammates in LLM course which is Large Language models
Which course me and Lizzy are partners?

Observation:

The chatbot correctly remembered that the user mentioned the LLM (Large Language Models) course and responded appropriately.

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Clear chat



My name is Priya and my friend name is Lizzy. we both project teammates in LLM course which is Large Language models



. We have been working together on various projects and assignments for the past few months. Lizzy is a great team player and always comes up with innovative ideas. I am grateful to have her as my project teammate. Together, we make a strong and efficient team, and I am confident that we will achieve great success in our LLM course.



which course me and lizzy are partners



You and Lizzy are partners in the LLM (Large Language Models) course.

Explanation:

This demonstrates that the chatbot maintains conversation history by sending all previous messages in `st.session_state.messages` to the API.

This confirms contextual memory is working correctly.

Technical Question Test

Test Prompt:

Explain overfitting. Can you give a simple example?

Observation:

The chatbot provided a structured explanation including:

- Definition of overfitting
- Example using polynomial regression
- Summary explanation



Explain overfitting.Can you give a simple example?



Overfitting occurs when a machine learning model learns the details and noise in the training data to the extent that it negatively impacts the model's performance on new, unseen data. In other words, the model performs very well on the training data but fails to generalize well to new data.

A simple example of overfitting can be seen in a linear regression model. Let's say you have a dataset with two features (X) and a target variable (Y). If you fit a linear regression model with a high degree polynomial (e.g., degree 10) to the data, it may perfectly fit the training data by capturing all the noise and fluctuations. However, this complex model may not generalize well to new data, leading to poor performance.

In summary, overfitting occurs when a model captures the noise and fluctuations in the training data instead of learning the underlying patterns, resulting in poor performance on unseen data.

Explanation:

This demonstrates the chatbot's ability to generate structured technical explanations with clarity and depth.



Write a short email asking for internship feedback.



Subject: Request for Internship Feedback

Dear [Recipient's Name],

I hope this email finds you well. I am writing to kindly request feedback on my internship experience with [Company Name]. As my internship comes to an end, I value your insights and suggestions for my professional growth.

I would appreciate it if you could take a few moments to share your feedback on my performance, strengths, areas for improvement, and any other comments you may have regarding my work during the internship. Your feedback is incredibly valuable to me as I continue to develop my skills and knowledge in this field.

Thank you in advance for your time and support. I look forward to hearing from you and learning from your feedback.

Best regards,

[Your Name] [Your Contact Information]

Multi-Question Prompt Test

Test Prompt:

Describe the gradient descent algorithm, explain it in simple terms and also how to cook an egg in bullet points?

Observation:

The chatbot successfully:

- Explained gradient descent technically
- Simplified the explanation
- Switched to a completely unrelated topic
- Formatted cooking steps in bullet points



Describe gradient descent, explain it in simple terms and also how to cook an egg in bullet points



Gradient descent is an optimization algorithm commonly used in machine learning and deep learning to minimize the loss function of a model. The goal of gradient descent is to iteratively update the parameters of a model in the direction of the steepest descent of the loss function, in order to reach the optimal set of parameters that minimizes the loss.

Here's a simple explanation of gradient descent:

1. Start by initializing the parameters of the model with random values.
2. Calculate the gradient of the loss function with respect to each parameter.
3. Update the parameters in the direction of the negative gradient to minimize the loss.
4. Repeat these steps iteratively until the loss function converges to a minimum.

Now, here's how to cook an egg in bullet points:

- Crack an egg into a bowl.
- Beat the egg with a fork until the yolk and white are well mixed.
- Heat a non-stick pan on medium heat.
- Add a little bit of butter or oil to the pan.
- Pour the beaten egg into the pan.
- Let the egg cook for a few minutes until the edges start to set.

Explanation:

This experiment demonstrates the chatbot's ability to:

- Handle compound prompts
- Switch domains within one response
- Follow formatting instructions
- Generate structured answers

Personal Advice:

Prompt: *How can I stay focused while studying?*

Response:

The chatbot provided structured, practical advice in numbered format. It suggested:

- Setting clear goals
- Creating a study schedule
- Minimizing distractions
- Using Pomodoro technique

- Staying organized
- Practicing mindfulness
- Maintaining physical health



How can I stay focused while studying?



Staying focused while studying can be challenging, but there are several strategies you can try to improve your concentration and productivity. Here are some tips to help you stay focused while studying:

1. Set specific goals: Define clear, achievable study goals and break them down into smaller tasks. This will help you stay motivated and focused on your progress.
2. Create a study schedule: Establish a study routine that includes dedicated study time for each subject or topic. A consistent schedule can help you develop a habit of studying regularly.
3. Minimize distractions: Identify and eliminate distractions in your study environment, such as noise, clutter, or electronic devices. Find a quiet and organized space where you can focus without interruptions.
4. Take breaks: Schedule short breaks during study sessions to rest and recharge your mind. Use techniques like the Pomodoro Technique, which involves studying for 25 minutes followed by a 5-minute break.
5. Stay organized: Keep your study materials and notes organized to easily access them when needed. Use tools like planners, notebooks, or digital apps to stay on track with your study plan.
6. Practice mindfulness: Stay present and focused on the task at hand by practicing mindfulness techniques such as deep breathing, meditation, or visualization.
7. Stay healthy: Take care of your physical and mental well-being by getting enough sleep, staying hydrated, eating nutritious foods, and exercising regularly. A healthy lifestyle can

Observations:

1. Structured Output:

The chatbot generated a well-organized numbered list, making the advice easy to read and actionable.

2. Practical Suggestions:

The response included real-world techniques such as the Pomodoro method, goal setting, and distraction control.

3. **Clarity and Tone:**

The explanation was clear, motivational, and suitable for a student audience.

4. **Generalization Ability:**

This demonstrates the chatbot's ability to handle non-technical, real-life questions beyond programming or machine learning topics.

Conclusion:

Overall, this assignment demonstrates how large language models can be integrated into interactive applications using Streamlit, enabling real-time conversational AI systems. The project highlights key concepts such as conversation history management, API integration, and response generation using OpenAI's GPT models.